

M. SHAHARYAR

AI Systems Engineer — RAG, LLM Pipelines & Real-Time Infrastructure

Remote (UTC+5, overlaps US/EU mornings) | shaharyar321321@gmail.com | shaharyar.dev | [GitHub](https://github.com/shaharyar) | [LinkedIn](https://www.linkedin.com/in/shaharyar)

SUMMARY

Solo engineer who ships production AI systems end to end: retrieval pipelines over multi-million-chunk corpora, agentic LLM tool loops, cost-engineered inference, and real-time collaboration infrastructure. TypeScript, Rust, and Python. Everything below is live or open source.

EXPERIENCE

NOVUS ATLAS — Founder & Engineer

Jun 2024 – Present · novusatlas.org · codex.novusatlas.org

Recall — production RAG research agent over published fiction corpora (live at recall.novusatlas.org):

- Built an agentic retrieval loop where the LLM searches, retrieves, and cites evidence through tools; hybrid retrieval fans out semantic (Qdrant), keyword (Meilisearch), and wiki candidates in parallel, then dedupes and reranks before the model sees results.
- Cut cost per deep-reasoning question from ~\$0.10 to \$0.04–0.06 via handle-based retrieval, progressive evidence disclosure, opt-in reasoning tokens, bounded tool loops, and token-budgeted conversation history.
- Designed the async ingestion pipeline (BullMQ): EPUB/wiki parsing → chunking → embedding (qwen3-embedding-8b) → Qdrant + Meilisearch indexing, with tiered retries, exponential backoff, and deterministic job IDs for dedup.
- Implemented usage-derived billing: per-step provider cost summed into credit charges over an idempotent Postgres ledger (advisory locks keyed on reference ID), so retried webhooks and duplicate jobs can never double-charge.

NOVA — Real-time collaborative writing platform:

- Engineered collaborative editing on Operational Transformation with presence awareness, autosave, optimistic synchronization, and offline-resilient recovery; sustained 500+ concurrent collaborators at <80 ms p95 sync latency.
- Built Rust-powered semantic search and NLP pipelines (vector embeddings, Qdrant, PostgreSQL) indexing 5M+ content chunks at <150 ms p95 retrieval.
- Shipped Git-style branch-based version control for fiction; platform infrastructure across Next.js, Hono/oRPC, Redis, BullMQ, Cloudflare R2, Socket.IO, and Rust/N-API.

KICKLAYER — Founder & Engineer - 2026 – Present · kicklayer.com

- Built a multi-tenant workflow platform that orchestrates client onboarding, asset collection, credential exchange, compliance tracking, and AI-assisted project preparation for agencies.
- Designed schema-driven onboarding workflows with conditional logic, progress tracking, file validation, version history, automated reminders, authenticated exports, and field-level audit trails.
- Built secure client infrastructure featuring passwordless authentication, AES-256-GCM credential storage, secret detection, access controls, token revocation, and activity tracking.
- Integrated AI-assisted asset verification, onboarding analysis, project brief generation, delay-risk detection, subscription enforcement, usage metering, and billing workflows.
- Reduced onboarding follow-up workload by 61% and shortened asset-collection timelines by 54%.

INTERPOL — Contract Data Engineer

- Built data-analysis pipelines over large investigative datasets (Python, pandas, PostgreSQL). Work under NDA; tooling and scale discussable on request.

SELECTED ENGINEERING PROJECTS

OpenOT — Open-Source Operational Transformation Framework · open-ot.shaharyar.dev

- Framework-agnostic OT engine with state-machine transforms, pluggable WebSocket/SSE transports, storage adapters, and React/Lexical integration kits.
- Correctness validated through 10,000+ transformation test cases; benchmarked at 150,000+ ops/sec (reproducible in repo).

The Architect Engine — AI-Native Narrative Platform · architect.novusatlas.org

- Built stateful LLM orchestration: server-side prompt architecture, typed JSON output contracts, retry pipelines, and model-routing strategies across providers.
- Adaptive story interviews with structured generation, branching scenario chains, and persistent state.

TECHNICAL SKILLS

AI/LLM: RAG architecture, agentic tool loops, vector search (Qdrant), hybrid retrieval, embeddings, structured generation, model routing, inference cost engineering, LLM orchestration

Languages: TypeScript, Rust, Python, SQL

Systems: Operational Transformation, distributed sync, WebSockets/SSE, event-driven systems (BullMQ, Redis), PostgreSQL, Cloudflare R2, Meilisearch

Product: React, Next.js, Node.js, Hono, tRPC/oRPC, Docker, Railway, Vercel

TECHNICAL WRITING

Architecture documentation and engineering deep-dives: codex.novusatlas.org